Group template: DH Project Statement, Results, & Data Management Plan

Weighted: 50%

Deadline: 2 December 2025

Note: Part of this template was adapted from the Science Europe Core Requirements for Data Management Plans.

Project Statement	
Project Title:	Exploring the impact the rate of AI adoption in the U.S. has on job loss in 2025
Author(s):	Ellen Higgins; Chunxiao Wang; Yilin Wen
Date Completed:	04/12/2025
Project Summary: The goal and purpose of the data collection/creation/ transformation processes (200 words)	The purpose of our data collection and transformation process was to investigate whether media sentiment toward AI aligns with actual industry trends in AI adoption and general unemployment rate in the U.S. in 2025. To answer this question, it is necessary to gather the quantitative data on AI adoption rates and unemployment rates from sources such as Impact of AI on Digital Media datasets from Kaggle and the U.S. Bureau of Labor Statistics. These datasets provided basic information for comparing the retail and marketing industries. Missing values for certain years were filled through linear interpolation and extrapolation to ensure a complete timeline for analysis. Cleaning the datasets and creating a dataset that focuses on our project is important. It helps us to narrow the scope without making the project too large. All transformations were recorded to maintain transparency. Filtering the English articles that were published in the US in 2025 and combining them into two PDF files: one on retail, one on marketing through ProQuest as text materials to analyze sentiment using Python and Voyant Tools. The text analysis allows us to research the positive and negative sentiment from those articles about AI in the retail and marketing industries.

Project Results

Note: that while this section may reflect components prepared for your group project outline, it should also represent the perspective of the completed project, which may have evolved.

1. Give a brief overview description of your completed project.
    - What was the goal for your project at the outset / what research question(s) did your work address?

The goal of our project was to investigate what kinds of sentiment that media narratives about AI in 2025 in the retail and marketing industries reflect, and whether media tone is similar to real employment trends in the U.S. in 2025. We aimed to research whether the widespread perception that AI causes significant job loss is supported by data or reinforced mainly through media reporting.

- Did you have an intended (or imagined) audience other than yourself and the module's instructional team?

  We imagined this work being useful to students and researchers interested in AI's impact on the labour market and media analysis. Additionally, practitioners in a specific industry that seeking a clearer understanding of whether public concerns about AI-driven job loss are supported by evidence or shaped by media reports.

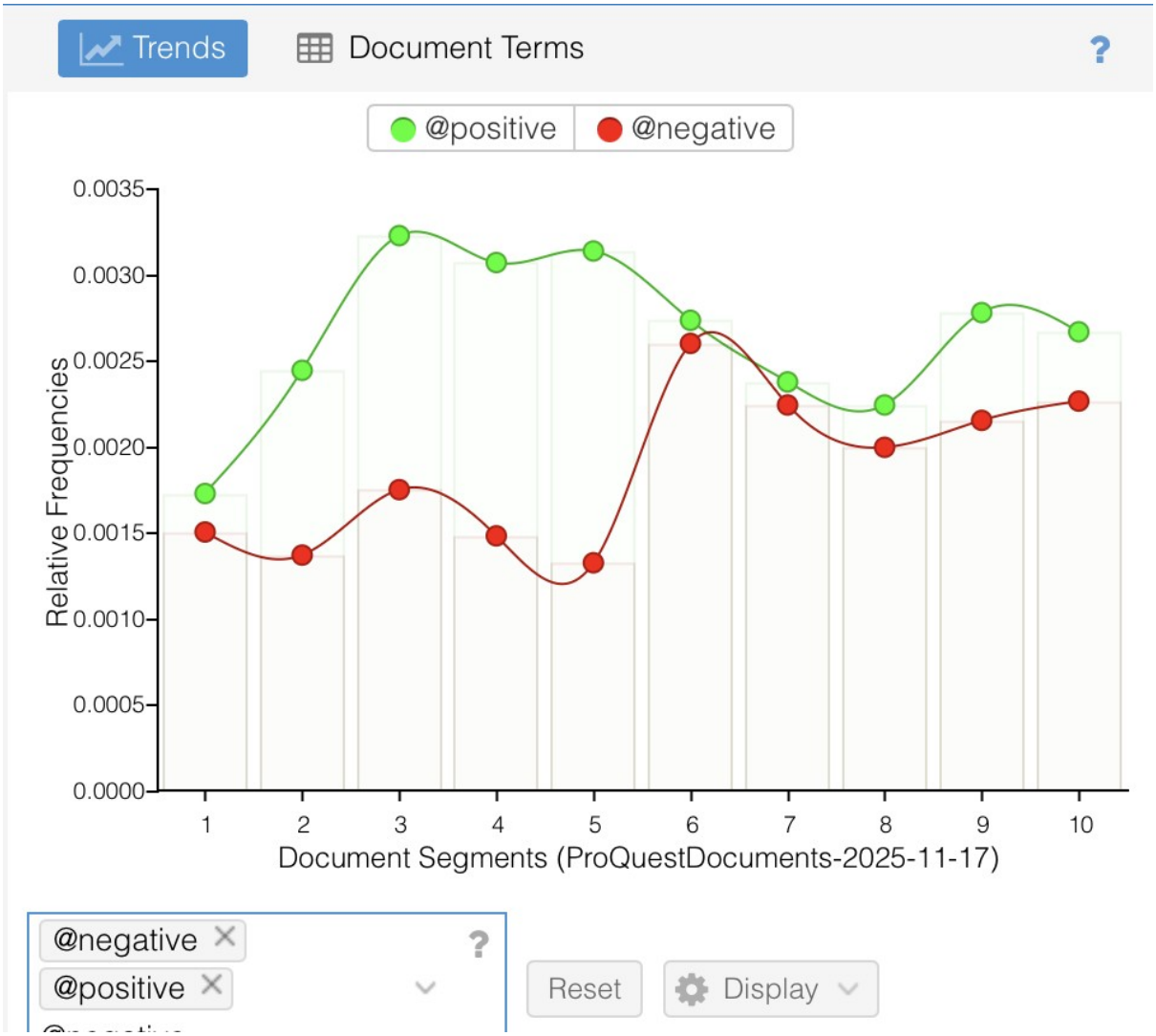
2. What tool(s) did you choose to meet this goal?
   Excel, ProQuest, Python(Jupyter notebook), Voyant Tools
   - What affordances and limitations did you keep in mind or discover when using this tool with your data?

     Affordances: Excel allowed us to analyze most of the numbers and visualize them in different charts. Using ProQuest ensured the sources were reliable, up-to-date, and relevant to our topics. We uploaded files into the Jupyter notebook, using matplotlib, because this library can analyze the data with customized visualizations that highlight key patterns. Voyant Tools offered quick text-analysis visualizations for comparing sentiment patterns, and it is accessible and user-friendly. At the same time, all of these visualizations are in different colours that strictly show the results.
     Limitations: The articles and reports we filtered in ProQuest are just a sample database and reflect partial aspects of our research question. Both Python and Voyant Tools rely on automated sentiment detection, which can misinterpret some context, leading to simplified sentiment scoring. Also, Python needs more technical skills to use.

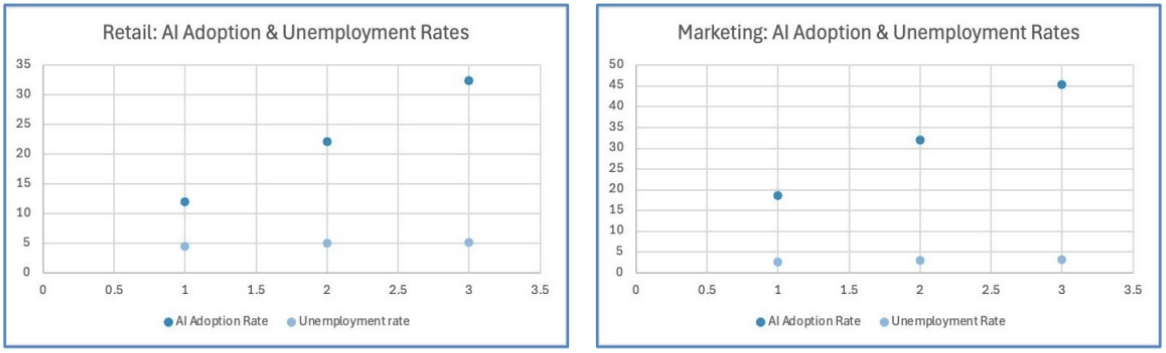
   - What did your choice of a digital approach allow you to do that an analogue approach could not have achieved?

     Using digital tools, we were able to clean, transform, and analyse large datasets efficiently, including showing interpolation and generating visualizations that would have been time-consuming by using an analogue approach. Additionally, digital tools provide access to multiple media articles and identify sentiment that would have required extensive manual reading and coding. The digital approach also ensured transparency and reproducibility in the process.

4. What form of output did you choose to meet your goals?
   - Include 1-3 key diagrammes or screenshots as applicable, and explain why you choose these visualisations.



This visualization clearly represents the positive and negative sentiment trends from articles and how the sentiment changes in a whole document. It is beneficial to reflect on the overall situation. As a complement to Python analysis.

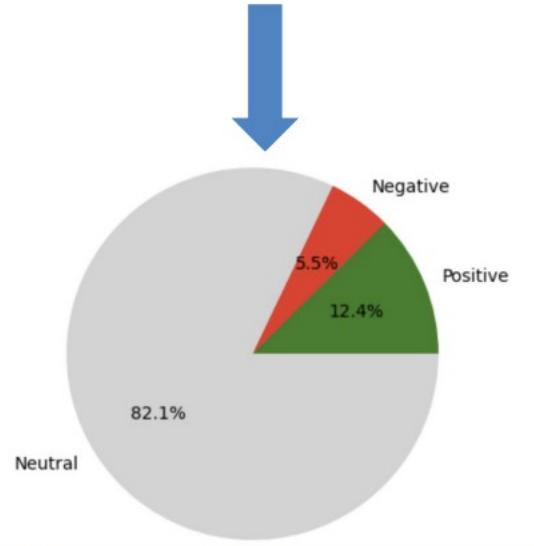


Through combined visualizations, placing both types of data on the same chart allows for a more direct comparison of the relationship between AI adoption rates and unemployment rates. Clearly shows readers whether the two curves move in the

same way. This provides visual evidence for the question of the relationship between AI adoption and job loss rates, making it clearer than viewing them separately.

Retail:

	neg	neu	pos	compound
0	0.055	0.821	0.124	1.0



Quantify media sentiment (positive/ negative/ neutral) to provide data-driven trends in media evaluation, rather than subjective feelings. The bar charts can quickly display the differences in the number of positive, negative, and neutral sentiments. Generating charts using code to make sentiment analysis results more accurate and reproducible.

- What affordances and limitations did you need to keep in mind in using this tool with your technology stack and audience requirements?

  Excel offered accessibility and transparency, making it easy for non-technical audiences to understand, but it has limited capacity for more complex analysis. Using Python provides efficient processing of large datasets, and it requires greater technical skill. ProQuest enabled access to credible media sources, but its filter system was restricted to the scope of articles we could include.

4. What would you consider to be your ‘results’?
   - What did you learn from the analysis, and how did you reach any conclusions?

     The rate of unemployment did not match the growth in AI use, and the adoption of AI did not significantly affect the unemployment rate, according to the U.S. media reports. Through visualizing the data about the AI adoption rate and unemployment rate in the U.S., and representing the sentiment proportion about articles, we can reach conclusions by comprehensively analyzing the two results.

   - Did these results meet the precise focus and scale of your research question?

The results differed somewhat from our expectations, but they indeed answered our research question.

- Why do you think this was or was not the case?

  I think this was the case because our dataset was limited, but sufficiently for trend comparison, and the combination of quantitative and textual analysis methods allowed us to present both industry data and media narratives within the specific scope.

Data Management Plan

1. **Data description and collection or re-use of existing data**
   a. What data (for example the kinds, formats, and volumes) will be collected or produced for the project (use the table below)?

In/Output #	Descriptor of the Digital Output	Type (e.g. file extensions)	Format//Size	Origin/Access Conditions (license, public domain, etc.)
1 Original Global AI Content Dataset	Countries, industries, time, and many indicators related to AI	.csv	15.8KB	Apache 2.0 open source license
2 Refined Global AI Content Dataset	Filtering the AI adoption rate in the U.S. from 2023 to 2025 (all industries)	.xlsx	55.8KB	Origin
3 Transformed Global AI Content Dataset	Filtering the AI adoption rate in the retail and marketing industry in the U.S. from 2023 to 2025. Filling the unemployment rate in the Excel. Visualizing AI adoption and unemployment.	.xlsx	64.8KB	Origin
4 ProQuest Documents Marketing	Combining the articles about AI in marketing in the U.S. in 2025	.pdf	2.46MB	Public domain

5 ProQuest Documents Retail	Combining the articles about AI in retail in the U.S. in 2025	.pdf	1.51MB	Public domain
6 Marketing Python Sentiment	Analysing sentiment in marketing through articles by Python	.ipynb	29.1KB	Origin
7 Retail Python Sentiment	Analysing sentiment in retail through articles by Python	.ipynb	29.2KB	Origin
8 Voyant Sentiment Marketing	Analysing sentiment in marketing through articles by Voyant Tools	.png	130KB	Origin
9 Voyant Sentiment Retail	Analysing sentiment in retail through articles by Voyant Tools	.png	127KB	Origin

b. How was the data collected (be specific)?

Firstly, we downloaded the dataset, titled “Impact of AI on Digital Media (2020-2025),” from Kaggle. The dataset was broad, and we narrowed the scope to the impact of AI on the U.S. retail and marketing from 2023 to 2025. There are some missing values in the dataset, so we used a data interpolation method to complete the dataset.
Then, to analyze the media narratives about AI-driven job loss, we filtered English articles that related to AI and job loss that were published in the U.S. in 2025 in ProQuest and combined those articles into PDF files: one on retail and one on marketing.

2. Documentation and data quality

a. What structural elements and/or metadata will accompany data? If you have structured data elements in your project, where did this structure come from?

Structural elements such as column headers (year, industry, AI adoption rate, unemployment rate), consistent file-naming conventions, and documented records of data cleaning steps. Metadata such as dataset source, author, license information, and filters used in ProQuest also accompany the data to ensure transparency.
Most of the structure came from the original datasets (Kaggle and the U.S. Bureau of Labor Statistics), which already provided clear variables. Additional structure, such as cleaned tables, combined datasets, was created during our processing in Excel and Python to standardize formats and make the data suitable for analysis.

b. What data quality control measures were taken? Did you have to clean or edit your input data in any way to make it usable? How did you document these changes, or at least make them reversible if required? If you did any data cleaning or processing or

choose not to, those processes should be described here in terms of decision making and software/tools used (*not step-by-step technical instructions*).

First, the raw datasets from Kaggle and the U.S. Bureau of Labor Statistics were duplicated to preserve the original files. We inspected the data in Excel to identify missing values, duplicated figures, and inconsistencies across years and industries. Then we use a data interpolation method to create a complete and logically consistent time series, documenting these changes in Excel.
All transformations, such as filtering, calculating, were recorded through saved Excel sheets, and different Excel versions to indicate the stage of processing. We used ProQuest filters consistently and recorded search parameters to control bias in media collection.

c. Are any standards being applied to the data? Consider your own data management (e.g., text mark-up used; file naming; or organisation of files)?

Yes, several data management standards were applied throughout the project. We used a consistent file naming format to distinguish raw, cleaned, and processed datasets. Files were organised into structured folders. Within Excel and Python, we maintained standardised column names. Documentation of data cleaning was retained in Jupyter Notebook comments. We also used a different color to highlight key patterns. These practices ensured reproducibility and consistency across the project.

3. Storage and data management responsibilities

a. How/where were data and metadata stored during the research process?

We downloaded data from Kaggle, the Bureau of the U.S. Labour Statistics, and ProQuest. Then we uploaded the datasets to OneDrive, and also stored the backup in the local documents.

b. Who was responsible for data management (i.e. the data steward)? How was this decided?

We each retain the data for our respective research sections, and uploaded it to our group OneDrive folder.

c. Comment upon the selected datasets and if/how they applied FAIR (Findable, Accessible, Interoperable, Re-usable) principles or not?

The datasets we used aligned with the FAIR principles, though with some limitations. The Kaggle dataset was highly Findable, with a clear URL, descriptive metadata, and searchable indexing on the platform. Both Kaggle and the U.S. Bureau of Labor Statistics datasets were Accessible, because they were available for public download in standard format. In terms of Interoperability, the use of common data structures, such as .csv files with column headings, .pdf files with clear instructions, made it easy to integrate them into Excel, Python, and Voyant Tools. The datasets were also Re-usable due to open licensing and transparent source documentation. However, some missing values and inconsistent reduced reusability without additional cleaning. ProQuest sources were less FAIR, as they were not openly accessible, and people needed to sign in through their institutions.

4. Legal and ethical requirements, codes of conduct

a. How were other legal issues, such as intellectual property rights and ownership managed? What legislation or copyright laws are applicable?

All data sources used in the project were licensed with intellectual property rights and licensing terms. The Kaggle dataset was distributed under the Apache 2.0 License, which explicitly allows reuse, modification, and redistribution as long as attribution is maintained. The U.S. Bureau of Labor Statistics data is part of the U.S. government public domain materials, and can be freely used with proper citation. For articles obtained through ProQuest, access was governed by the university's institutional subscription, and all text citations remained within the boundaries of licensed access.

The main legislation is the Copyright and Related Rights Act 2000, which has been updated by the Copyright and Other Intellectual Property Law Provisions Act 2019. These laws are heavily influenced by EU directives, particularly the Copyright Directive 2001/29/EC and the Enforcement Directive 2004/48/EC. The 2019 Act modernizes previous laws and expands court access for infringement claims, while also implementing changes regarding digital content and educational exceptions.

b. How were possible ethical issues be taken into account, and codes of conduct followed?

Possible ethical issues were included to ensure that all data used in the project came from publicly available and legally accessible sources. We maintained transparency in data cleaning and avoided bias that could misrepresent trends. We followed academic codes of conduct by citing all datasets and materials accurately and using them only for educational purposes.

c. If personal data were processed, how was compliance with legislation on personal data and on data security ensured (in particular GDPR)?

No personal data was processed in this project. All datasets consisted of compiled industry statistics and publicly published media articles.

5. Data sharing and long-term preservation

a. Will data be shared? How and when? Are there possible restrictions to data sharing or embargo reasons?

The cleaned and processed datasets produced for this project can be shared within our group coursework, but they will not be publicly released.
The media articles obtained through ProQuest are subject to subscription access and can not be redistributed. The datasets from Kaggle and the U.S. Bureau of Labor Statistics are openly available, so their original sources can be shared instead of reposting the files.
No embargo periods are required, but sharing needs to exclude the copyrighted media text.

b. How will your project, or parts of it, be sustained? For how long? After that point, how will it be de-commissioned?

The project and its core materials, such as cleaned datasets, analysis notebooks, and visualizations, will be sustained for as long as required for potential assessment, stored securely in OneDrive cloud storage and local backups. After the necessary academic period, the project will be decommissioned by archiving the working files and removing ProQuest materials with institutional policies that cannot be retained beyond educational use due to licensing restrictions.

c. What methods or software tools will be needed to access and use the data? Do these pose any sustainability risks?

Accessing and using the data requires software tools such as Excel, Python (especially Jupyter Notebook), and data visualisation platforms like Voyant Tools. These tools are widely used, well-supported, and unlikely to break down in the short term. The potential risk comes from relying on external platforms, such as ProQuest and Voyant Tools, because long-term access depends on institutional subscription or continued platform maintenance. However, the core datasets, like CSV files and PDF files, remain accessible independently of these tools, ensuring that the essential data can still be used even if specific platforms become unavailable.

References

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